Module 4: Intermediate HTML and CSS

Semantic HTML5 Elements

- Semantic tags, like HEADER and FOOTER, help make web pages clearer for both users and search engines, enhancing accessibility and SEO.
- Using semantic tags improves accessibility for users with disabilities by assisting screen readers in navigating content. They also help search engines index and rank pages more effectively, improving site visibility.
- Benefits of semantic tags include better code readability, easier maintenance, and a clear content structure. For example, a layout without semantic tags is less clear than one using them. Key semantic HTML5 elements include HEADER for introductory content, NAV for navigation links, SECTION for defining sections, ARTICLE for self-contained content, ASIDE for related content, and FOOTER for document footers.
- To use semantic tags effectively, choose tags that accurately represent the content, maintain consistency across your project, and validate your HTML with tools like the W3C HTML Validator.

Advanced CSS Selectors

- Attribute selectors allow styling of elements based on attributes, such as targeting a span with a CLASS or styling input elements with a TYPE of SUBMIT.
- Pseudo-elements enable styling specific parts of elements, like the first letter or line, and allow the addition of content before or after elements.
- Pseudo-classes style elements based on their state or position, with examples including modifying hyperlink styles on hover and targeting first child elements.
- Relationship selectors select elements based on their DOM position, targeting direct children or descendants within a hierarchy.

Generating Basic Layout using CSS

- Block elements (like DIV and h1) take full width and start on a new line, while inline elements (like SPAN and A) only use necessary width.
- The POSITION property controls layout: STATIC positions elements normally, RELATIVE adjusts positions relative to their original spot, ABSOLUTE positions them relative to the nearest positioned ancestor, and FIXED keeps elements in a fixed place on the browser window.
- The FLOAT property allows elements to stack left or right, useful for layouts with text wrapping around images. The CLEAR property can break floating stacks by preventing elements from appearing next to floated elements.
- Dimensions can be defined using absolute units (like pixels) or relative units (like EM and VW). The BOX-SIZING property helps control total element dimensions by adjusting how width and height are calculated.

CSS Flexbox

- CSS Flexbox is a layout model for arranging items in a single dimension (row or column) using a flex container and its flex items. The flex container is created by setting the display property to flex.
- Key Properties of Flex Container:
    - flex-direction: Determines the direction of flex items.
    - justify-content: Distributes space along the main axis.
    - align-items: Distributes space along the cross axis.
    - flex-wrap: Controls whether flex items should wrap.
- Flex containers are defined in CSS with display: flex, and items are arranged in a row by default.
- The flex-direction property controls the arrangement of items, allowing options like row, row-reverse, or column. This property manages space distribution on the main axis, with values such as flex-start, flex-end, centre, space-between, space-around, and space-evenly.
- The cross axis is perpendicular to the main axis, with align-items controlling space distribution along it.
- The default is stretch, but it can also align items to the end of the cross axis.
- This property controls how flex items grow relative to each other, allowing for dynamic sizing based on specified values.

CSS Grid

- CSS Grid is a two-dimensional layout system, unlike Flexbox, which is one-dimensional. A Grid Container holds Grid Items, which can span multiple rows and columns.
- To enable CSS Grid, set the display property to grid and define columns with the grid-template-columns property. The rows automatically adjust to content with grid-template-rows set to auto. Gaps between items can be controlled using the grid-gap property.
- CSS Grid also allows defining specific areas within the grid using grid-template-areas.
- For multiple columns, the repeat function can be used. Items can span across rows or columns using the grid-column and grid-row properties. This flexibility makes CSS Grid a powerful tool for creating complex, responsive layouts.

Introduction to Responsive Design

- Responsive design ensures web pages display well on various devices and screen sizes, adapting layouts for optimal viewing on desktops, tablets, and smartphones.
- For example, on a large screen, a website might use a three-column layout. On a tablet, it adjusts to two columns, allowing better visibility for elements like maps. On smartphones, it simplifies to one column to ensure easy viewing.
- Responsive design enhances user experience, increases reach due to rising mobile use, and benefits search engine optimisation by improving site visibility. Key components include fluid design, which uses percentages for flexible layouts; flexible images, which scale appropriately; and media queries, which apply styles based on device characteristics.
- Testing responsive design is essential. Use browser developer tools to simulate different screen sizes.

Working with Frontend Frameworks and Templates

- Frameworks standardise development, enhancing efficiency for frontend and backend systems. In frontend development, two main types exist: JavaScript frameworks for systematic coding and CSS frameworks like Bootstrap, which provide pre-built styles and UI components.
- Bootstrap is the most popular frontend library for mobile-first sites. Setup involves including the necessary CSS and JavaScript files in your project, and a sample HTML snippet is provided to test your setup. The Bootstrap grid system uses a 12-column layout, adaptable to different screen sizes, allowing for responsive designs.
- You can define equal-width columns or specific widths, with classes adjusting based on screen size. Bootstrap also includes predefined styles for text and forms, enhancing visual consistency. Responsive navigation elements and templates built on Bootstrap are available for customisation.

THE END

My notes...